

Proposed Research Title

An Explainable and Mobile-Deployable Deep Learning Framework for Diabetic Foot Ulcer Detection Using Smartphone Images

1. Background and Motivation

Diabetic Foot Ulcer (DFU) is a major complication of diabetes and a leading cause of lower-limb amputation worldwide. Early detection through visual inspection can significantly reduce severe outcomes. With the increasing availability of smartphones, mobile-based remote DFU screening systems offer a cost-effective solution, especially in low-resource and rural settings.

Existing deep learning approaches for DFU detection primarily focus on binary classification using clinical images. However, most of these systems operate as black boxes, lack explainability, and do not address real-world challenges such as variable image quality, uncertainty in predictions, and suitability for mobile deployment.

2. Research Problem

Most current DFU detection studies using public datasets focus only on classification accuracy and do not:

- Provide visual explanations to support clinical trust
- Assess model confidence or uncertainty
- Consider real-world mobile image variability
- Evaluate suitability for lightweight, on-device deployment

This limits their applicability in real telemedicine scenarios.

3. Research Objectives

The objectives of this project are:

1. To develop a deep learning model for **automatic DFU detection** using smartphone images.
2. To integrate **explainable AI (XAI)** techniques to visually highlight ulcer-relevant regions.
3. To estimate **prediction confidence/uncertainty** to improve clinical reliability.

4. To optimize the model for **mobile deployment** with low computational cost.
 5. To evaluate model robustness under real-world imaging conditions.
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4. Dataset

This research will use the publicly available Kaggle dataset:

Diabetic Foot Ulcer (DFU) Dataset

<https://www.kaggle.com/datasets/laithjj/diabetic-foot-ulcer-dfu>

Dataset Characteristics:

- Contains images of **healthy feet and diabetic foot ulcers**
 - Captured using **smartphone devices**
 - Includes variations in lighting, background, and angle
 - Suitable for real-world telemedicine research
 - Binary labels: Normal vs Ulcer
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5. Methodology

Step 1: Data Preprocessing

- Image resizing and normalization
 - Data augmentation (rotation, brightness, contrast)
 - Train-validation-test split
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Step 2: DFU Detection Model

- Use transfer learning with lightweight CNNs such as:
 - EfficientNet-B0
 - MobileNetV2
 - Fine-tune final layers for binary classification
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Step 3: Explainable AI Integration

- Apply Grad-CAM or Layer-CAM to generate heatmaps
- Visualize ulcer-relevant regions influencing predictions

- Compare explainability outputs across models
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Step 4: Uncertainty Estimation

- Implement Monte-Carlo Dropout during inference
 - Output prediction confidence alongside class label
 - Identify low-confidence cases requiring medical consultation
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Step 5: Mobile Optimization

- Convert trained model to TensorFlow Lite
 - Apply model quantization
 - Evaluate inference time and model size
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Step 6: Evaluation

- Metrics:
 - Accuracy, Precision, Recall, F1-score
 - ROC-AUC
 - Qualitative evaluation of explainability maps
 - Robustness testing under varying image conditions
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6. Novelty and Contribution

This work improves upon existing DFU detection studies by:

- Introducing **explainable AI** to improve clinical trust
- Incorporating **prediction uncertainty**, which is rarely addressed
- Focusing on **real-world smartphone deployment**
- Evaluating robustness under non-clinical imaging conditions

Unlike many prior works that focus solely on classification accuracy, this research aims to develop a **clinically interpretable and deployment-ready system**.

7. Expected Outcomes

- A lightweight DFU detection model suitable for mobile devices
 - Visual explanation of predictions for clinician and patient trust
 - Confidence-aware predictions for safer medical decision support
 - A foundation for future integration into mobile health applications
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8. Significance of the Study

This project contributes toward accessible, explainable, and reliable AI-based healthcare solutions. It is particularly relevant for remote and resource-limited settings, aligning well with modern telemedicine needs.